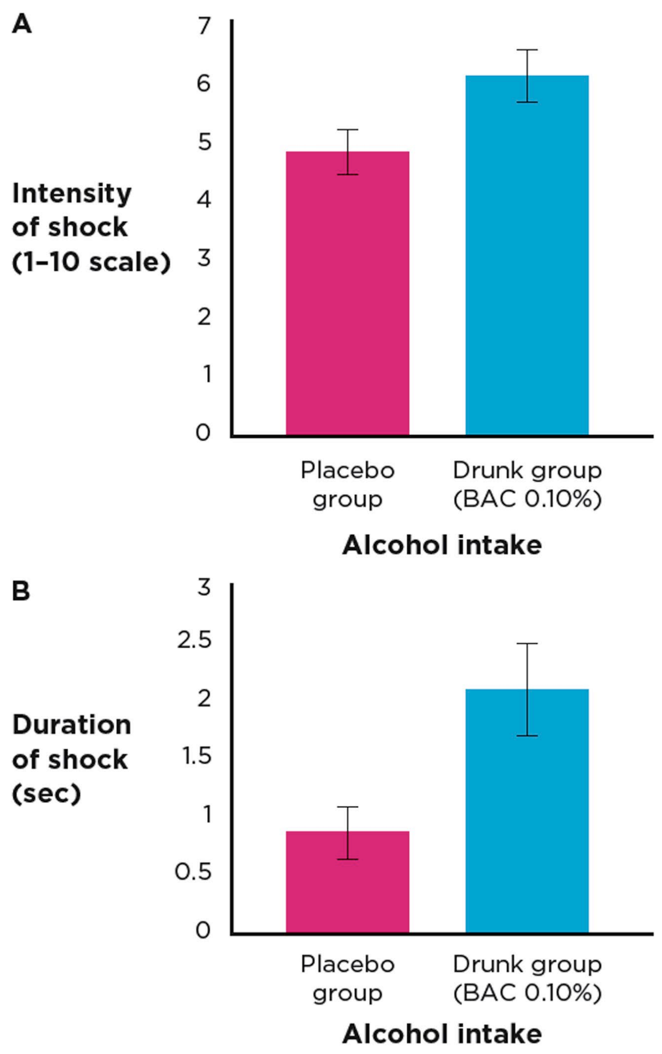
Experimental Design

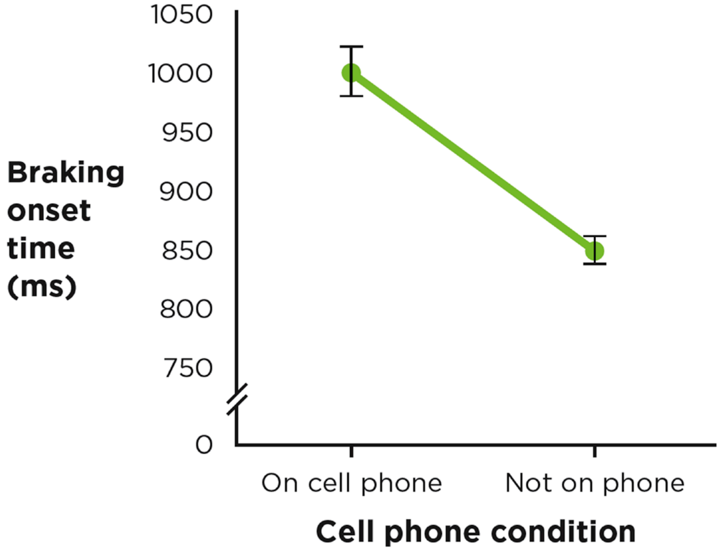
Week 13

Roadmap

- Review: Experiments with one independent variable
- Experiments with two independent variables can show interactions
- Factorial variations
- Identifying factorial designs in your readings

Review: Experiments with One Independent Variable



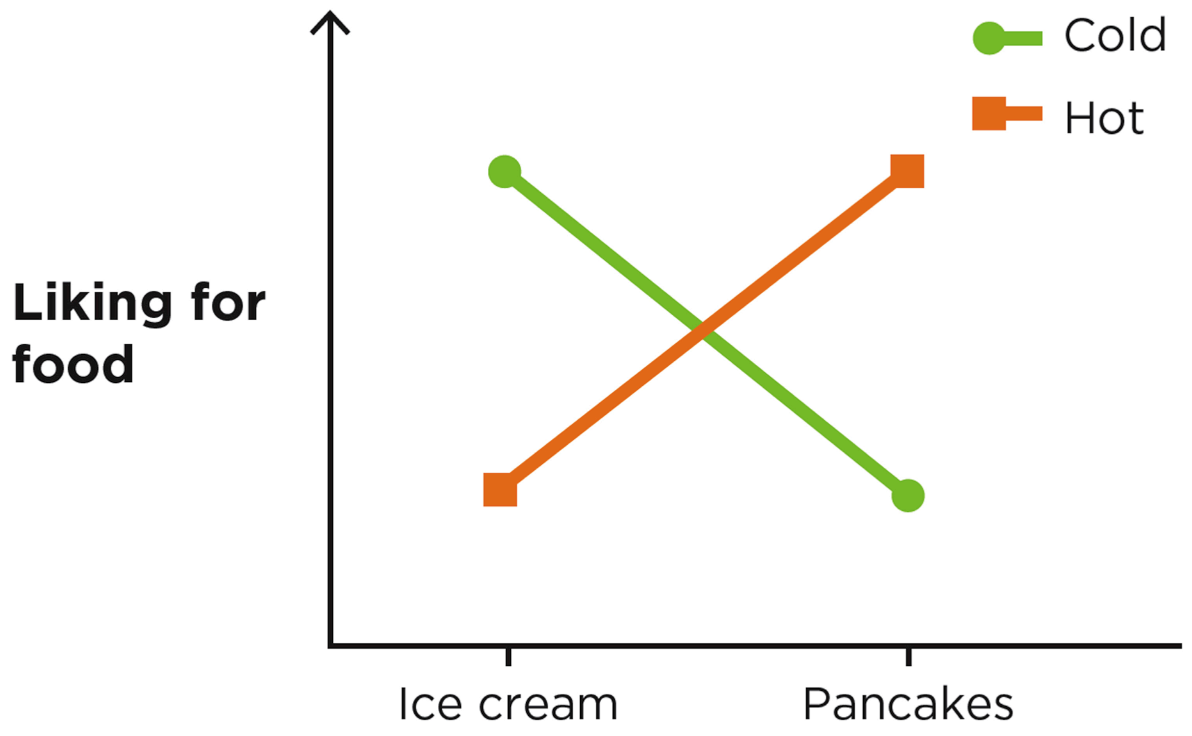


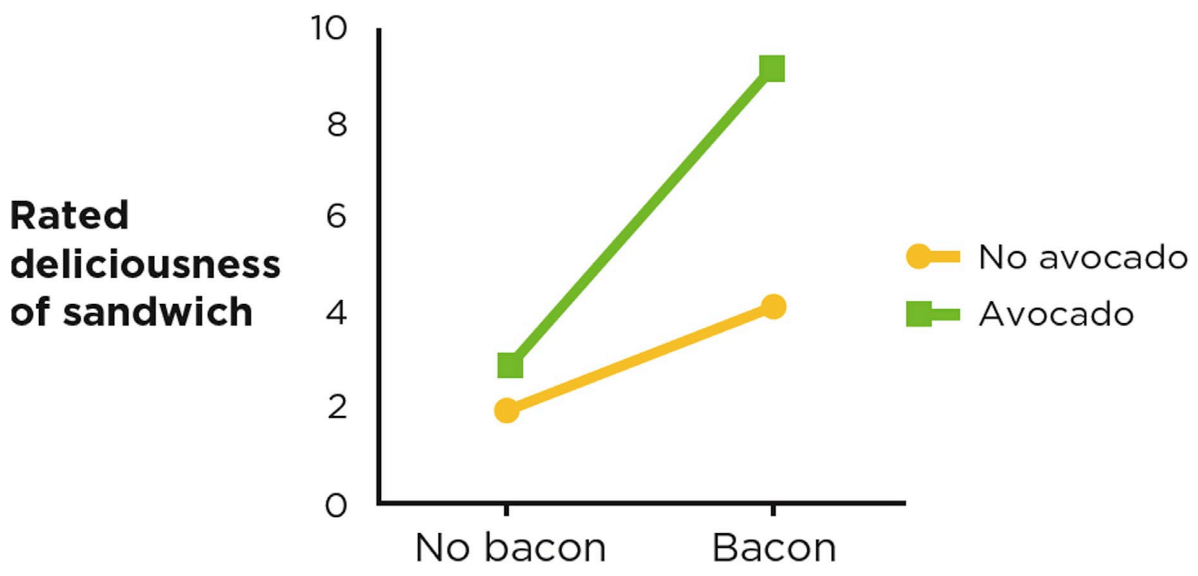
Experiments with Two Independent Variables Can Show Interactions

Interaction: Definition

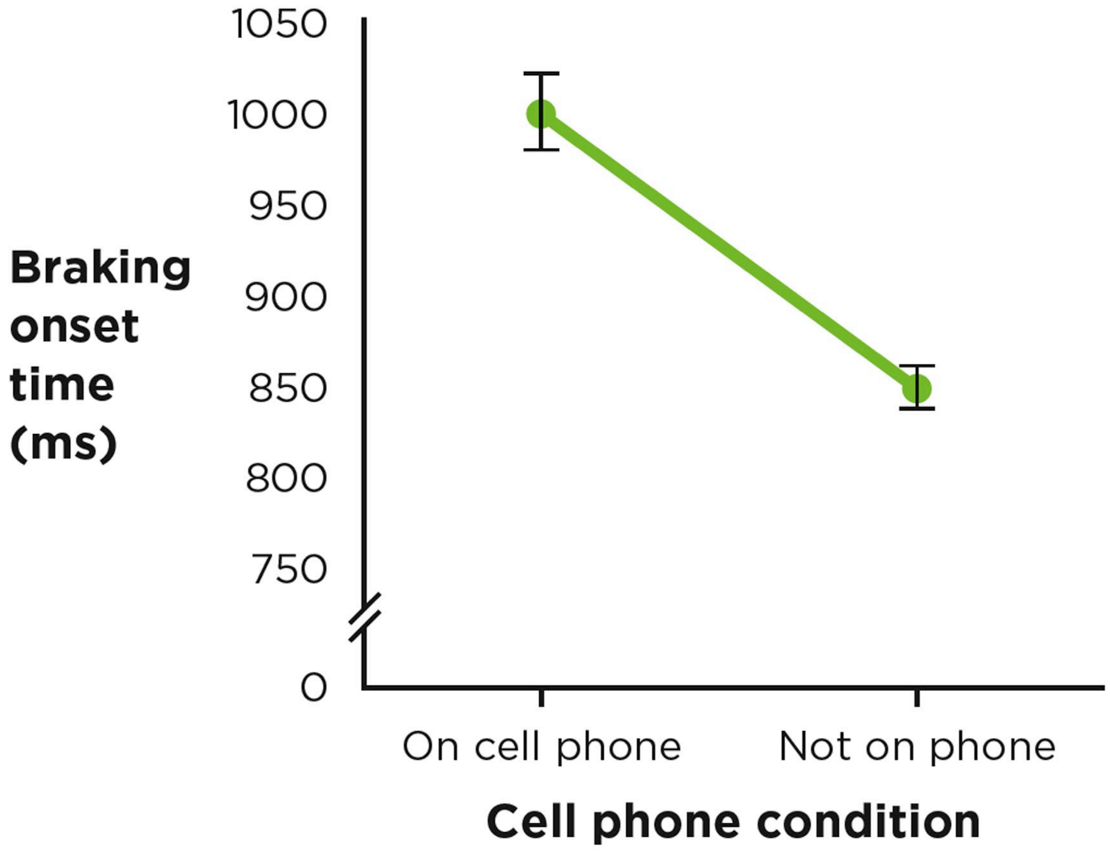
- Interaction =
- A difference in differences =
- The effect of one independent variable depends on the level of the other independent variable

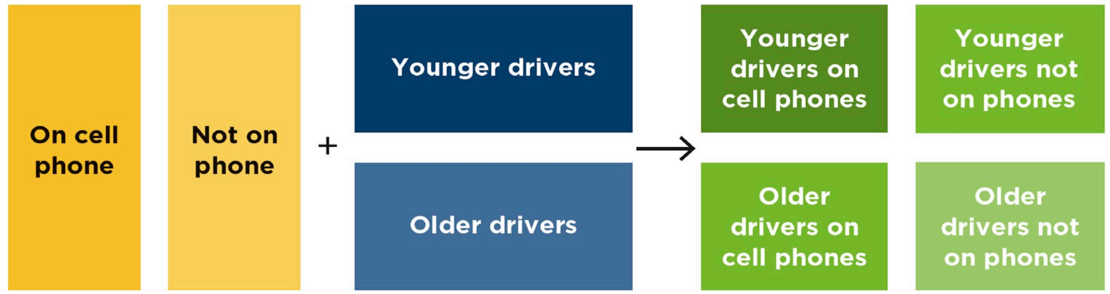
Intuitive Interactions



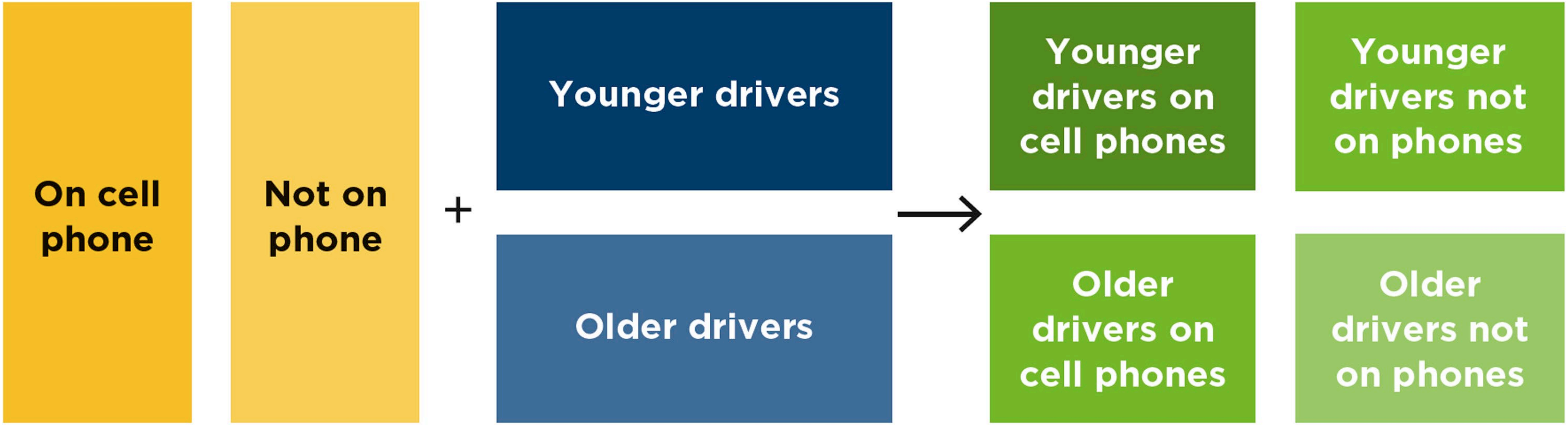


Reconsidering the cell phone and driving study





Factorial Designs Study
Two Independent Variables

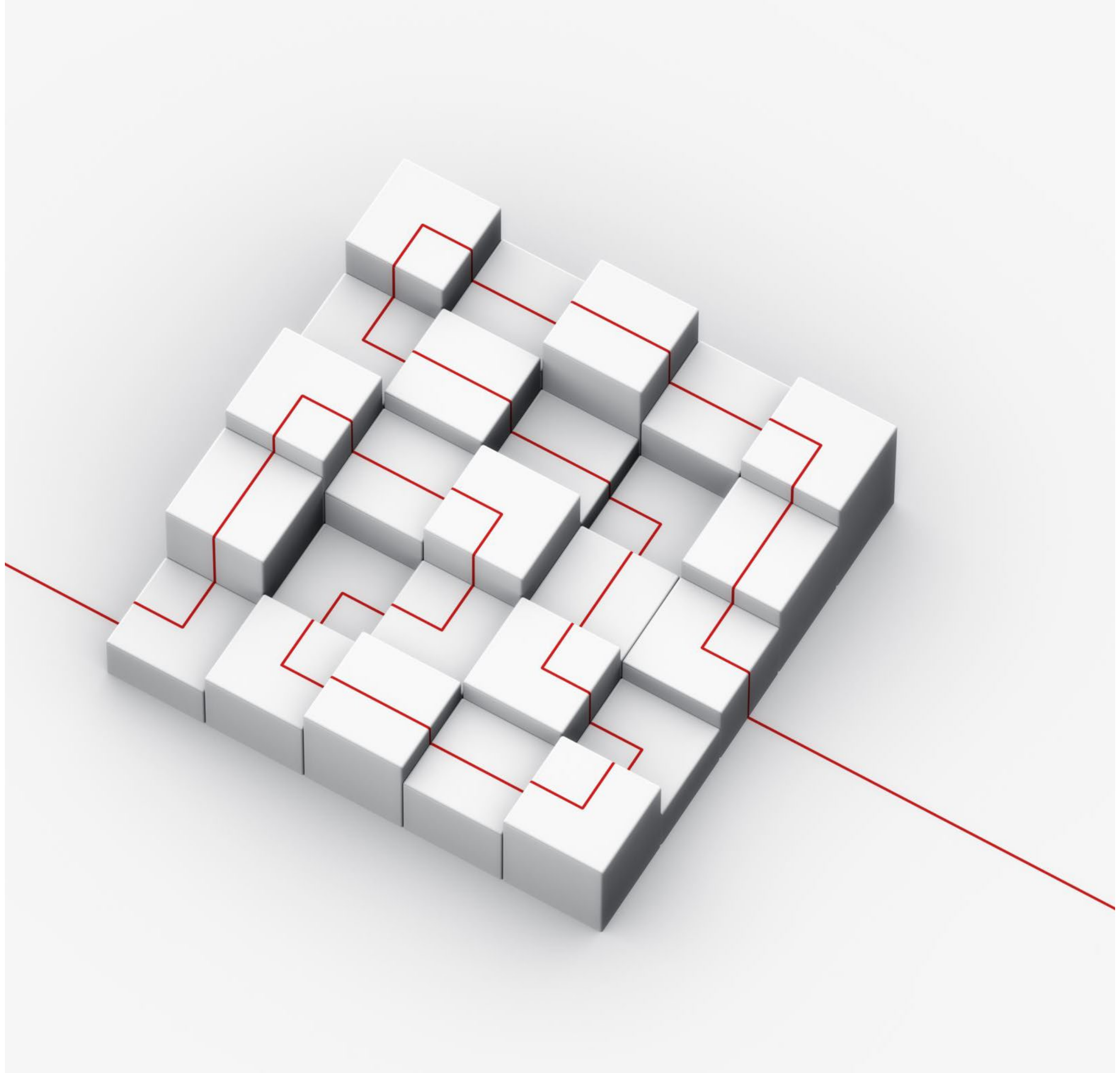


Factorial Designs Study Two Independent Variables

- **Factorial design**
- **Cell**
- **Participant variable**

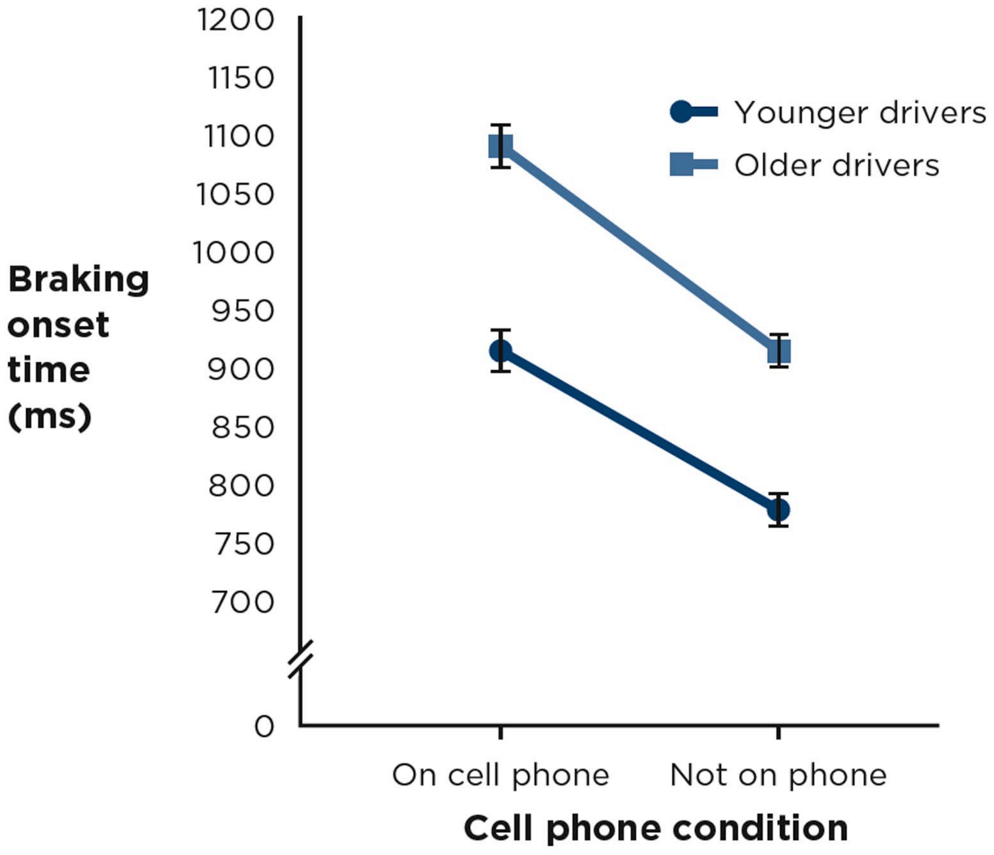
Factorial Designs Can Test Limits

- Sometimes, researchers conduct factorial designs to test
- whether an IV affects
  - different kinds of people or
  - people in different situations
  - in the same way.



Factorial Design Results in Table and Graph Formats

DV: Braking onset time (ms)		IV$_1$: Cell phone condition	
		On phone	**Not on phone**
IV$_2$: Driver age	**Younger drivers**	**912**	**780**
	Older drivers	**1086**	**912**



Factorial Designs Can Test Limits (continued)

- A form of external validity
- Interactions show moderators

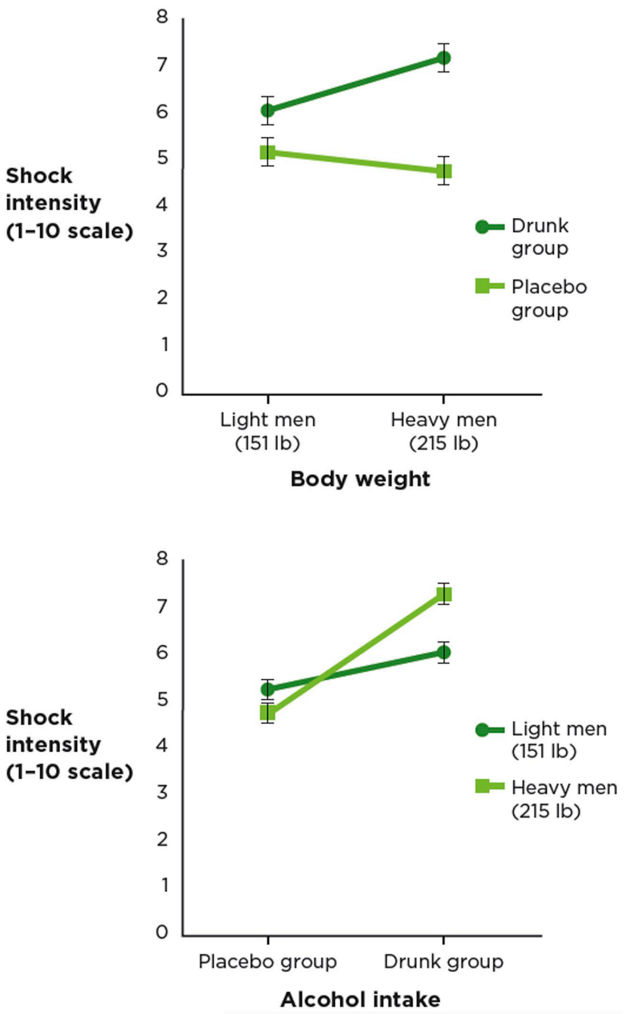
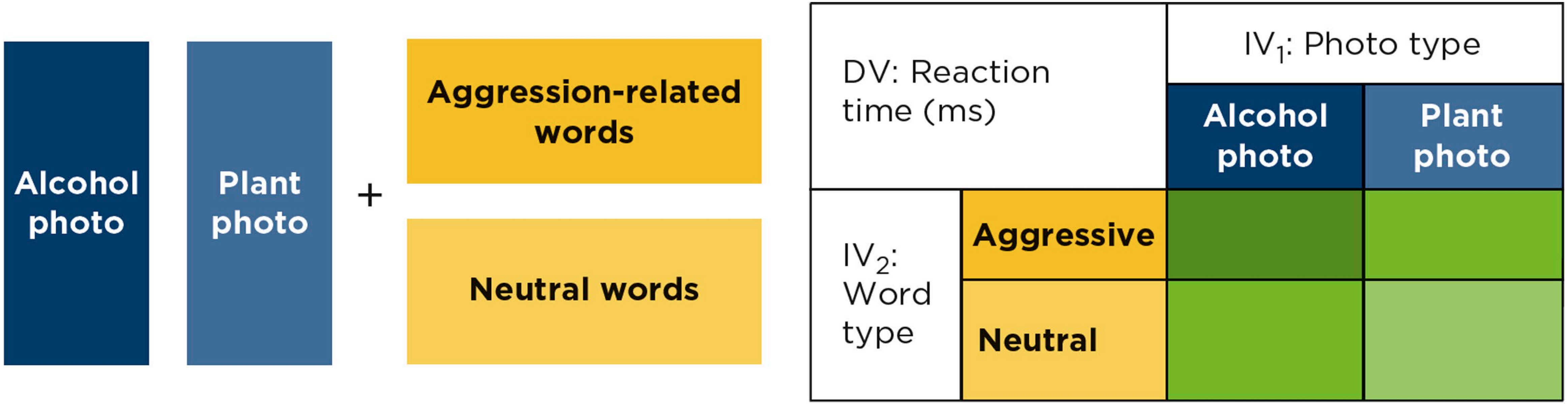


FIGURE 12.8

Factorial Designs Can Test Theories

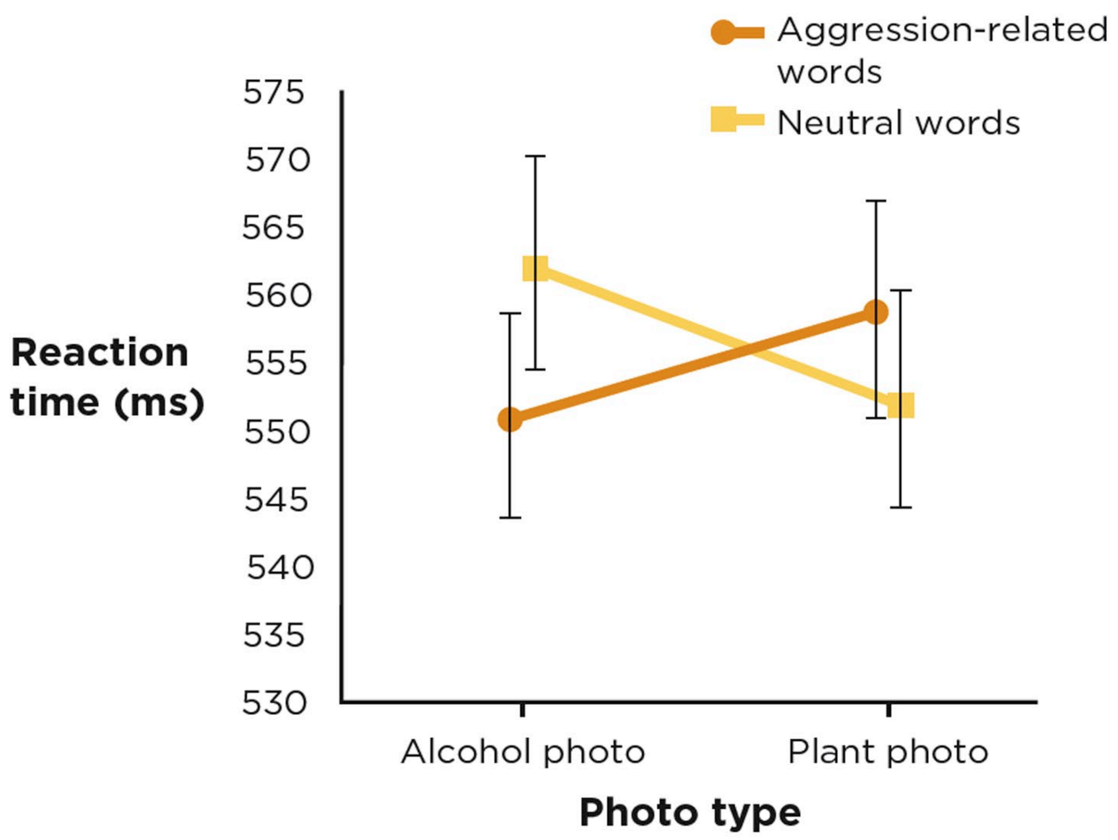
- Using a factorial design to test a theory of alcohol cues
- Using an interaction to test a memory theory

Using a Factorial Design to Test a Theory of Alcohol Cues



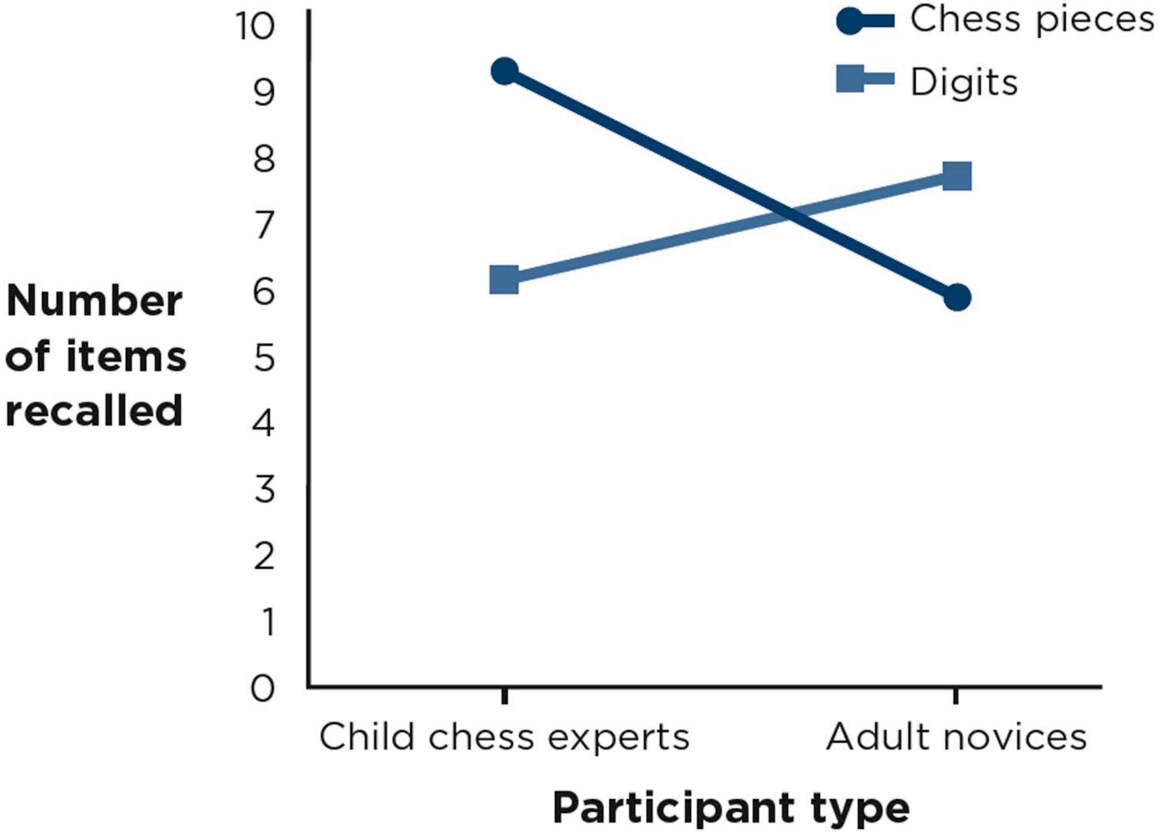
Using a Factorial Design to Test a Theory of Alcohol Cues: Results

DV: Reaction time (ms)		IV$_1$: Photo type	
		Alcohol	**Plant**
IV$_2$: Word type	**Aggressive**	**551**	**559**
	Neutral	**562**	**552**



Using an Interaction to Test a Memory Theory

DV: Items recalled		IV$_1$: Participant type	
		Child chess experts	**Adult novices**
IV$_2$: Type of item	**Chess pieces**	**9.3**	**5.9**
	Digits	**6.1**	**7.8**



Interpreting Factorial Results: Main Effects and Interactions

Main effect: Is there an overall difference?

Interactions

- Is there a difference in differences?
- More important than main effects

Possible main effects and interactions in a 2 x 2 factorial

Main Effect: Is There an Overall Difference?

DV: Reaction time (ms)		IV$_1$: Photo type		
		Alcohol	**Plant**	Main effect for IV$_2$: Word type
IV$_2$: Word type	**Aggressive**	**551**	**559**	**555 (average of 551 and 559)**
	Neutral	**562**	**552**	**557 (average of 562 and 552)**
Main effect for IV$_1$: Photo type		**556.5 (average of 551 and 562)**	**555.5 (average of 559 and 552)**	

DV: Shock intensity (1–10 scale)		IV$_1$: Body weight		
		Light men (151 lb)	**Heavy men (215 lb)**	Main effects for IV$_2$: Drinking condition
IV$_2$: Drinking condition	**Placebo group**	**5.09**	**4.72**	**4.91 (average of 5.09 and 4.72)**
	Drunk group	**6.00**	**7.19**	**6.60 (average of 6.00 and 7.19)**
Main effect for IV$_1$: Body weight		**5.55 (average of 5.09 and 6.00)**	**5.96 (average of 4.72 and 7.19)**	

Interactions: Is There a Difference in Differences? Options 1 and 2

DV: Reaction time (ms)		IV$_1$: Photo type	
		Alcohol	**Plant**
IV$_2$: Word type	**Aggressive**	**551**	**559**
	Neutral	**562**	**552**

Difference is –11 (551 – 562 = –11)

Difference is 7 (559 – 552 = 7)

Differences are statistically significantly different (–11 < 7). There is an interaction.

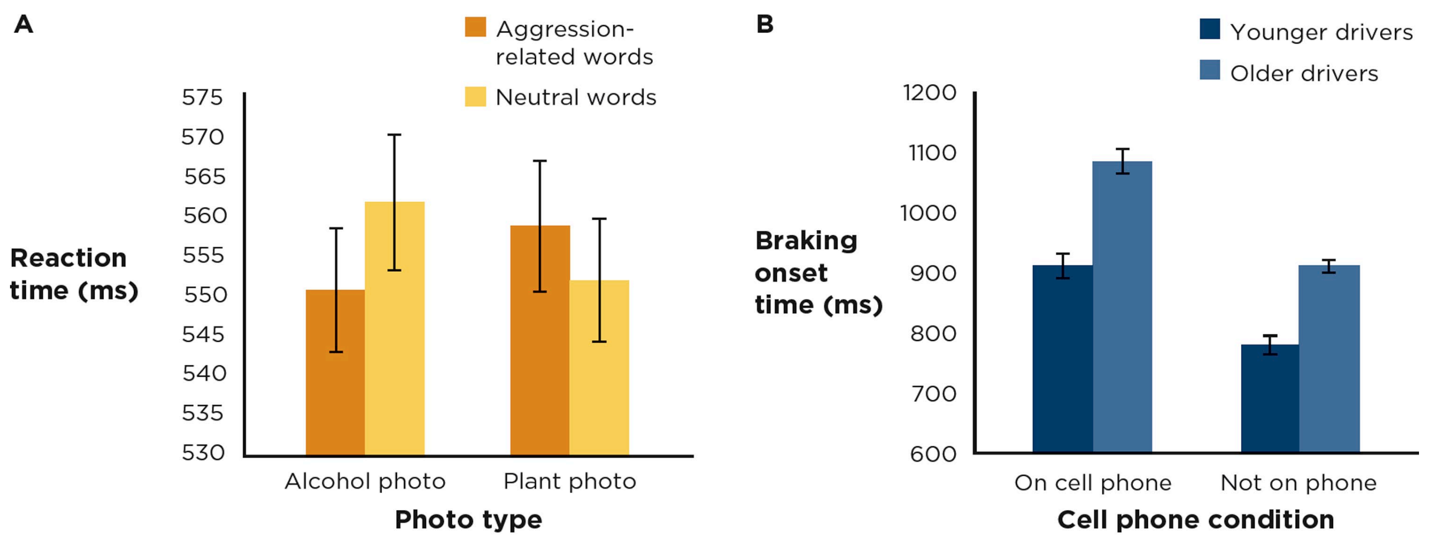
DV: Braking onset time (ms)		IV$_1$: Cell phone condition	
		On cell phone	**Not on phone**
IV$_2$: Driver age	**Younger drivers**	**912**	**780**
	Older drivers	**1086**	**912**

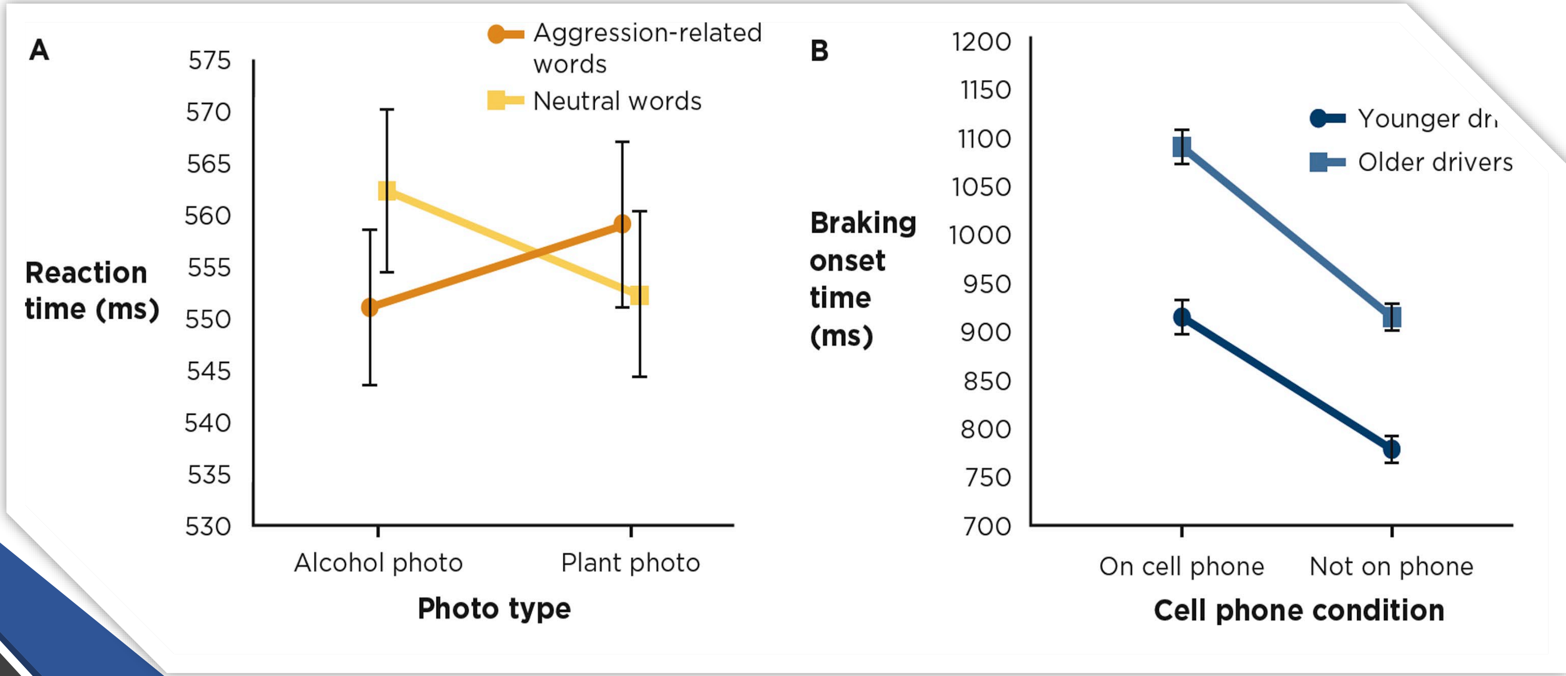
Difference is 132 (912 – 780 = 132)

Difference is 174 (1086 – 912 = 174)

These differences are not significantly different (174 is not significantly larger than 132). There is no interaction.

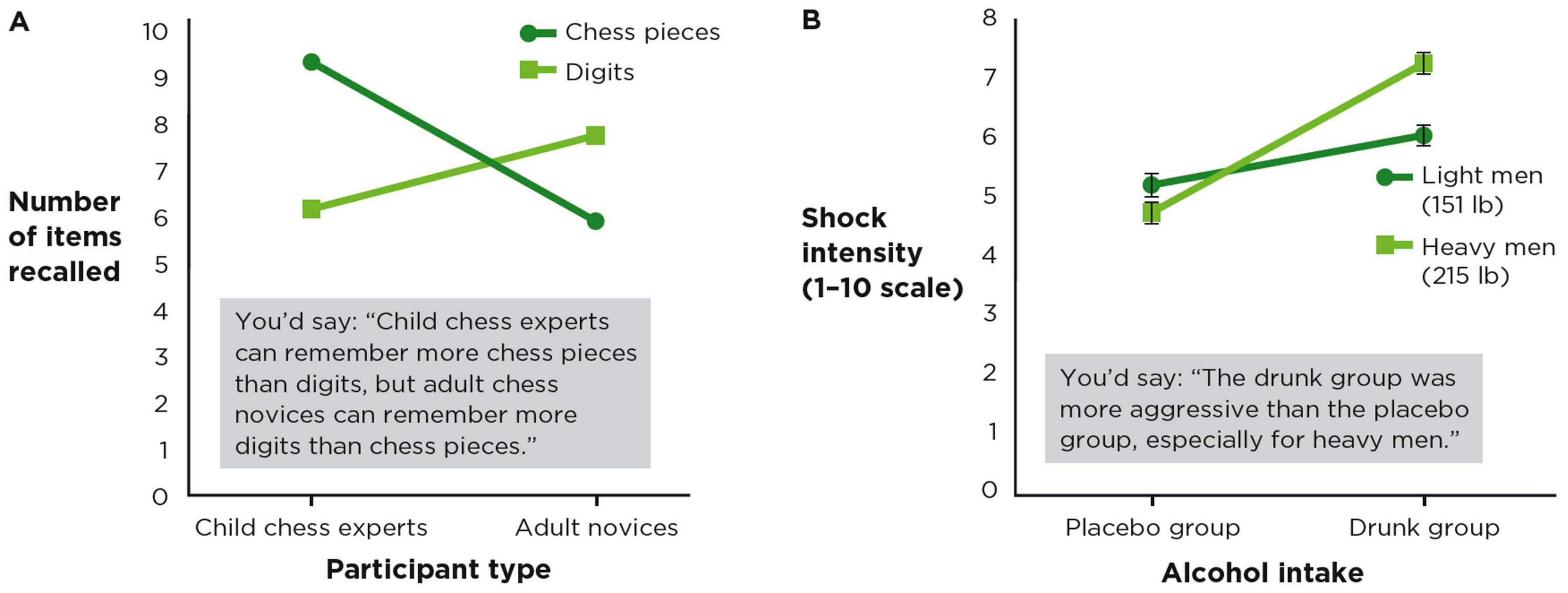
Interactions: Is There a Difference in Differences? Line and Bar Graphs





Interactions: Is There a Difference in Differences? Words

- Describing interactions in words



Interactions Are More Important Than Main Effects

When a study shows both a main effect and an interaction,

- the interaction is almost always more important.

There may be real differences in marginal means,

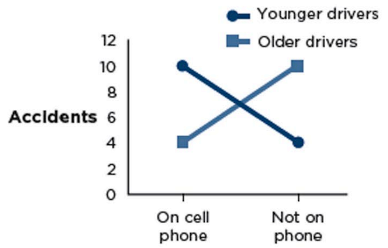
- but the more exciting part is the interaction.

Possible Main Effects and Interactions in a 2 x 2 Factorial Design

Summary of effects	Cell means and marginal means	Line graph of the results

Main effect for cell phone: ***No***
Main effect for age: ***No***
Age × cell phone interaction: ***Yes***—younger drivers on a cell phone cause more accidents; older drivers on a cell phone cause fewer accidents

DV: Number of accidents	On cell phone	Not on phone	
Younger drivers	10	4	7
Older drivers	4	10	7
	7	7	



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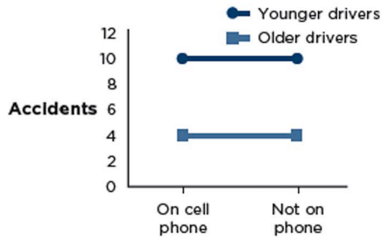

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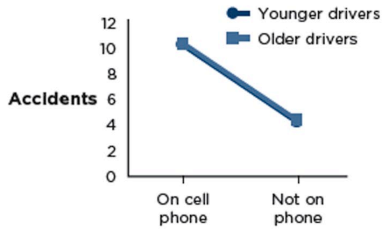
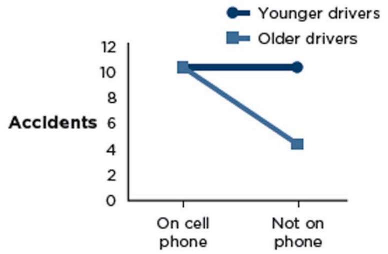


Main effect for cell phone: ***Yes***—cell phones cause more accidents
Main effect for age: ***Yes***—younger drivers have more accidents
Age × cell phone interaction: ***Yes***—for younger drivers, cell phones do not make a difference, but for older drivers, cell phones cause more accidents

DV: Number of accidents	On cell phone	Not on phone	
Younger drivers	10	10	10
Older drivers	10	4	7
	10	7	



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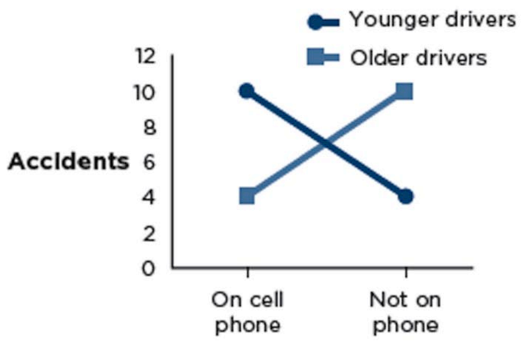

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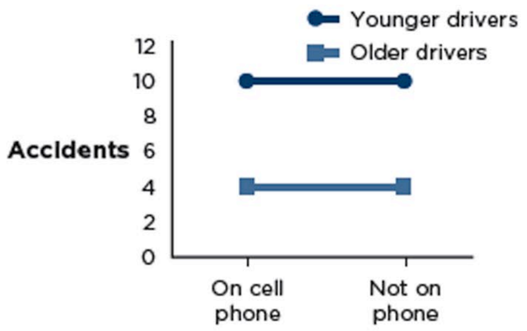

Possible Main Effects and Interactions in a 2 x 2 Factorial Design

Main effect for cell phone: ***Yes***—cell phones cause more accidents

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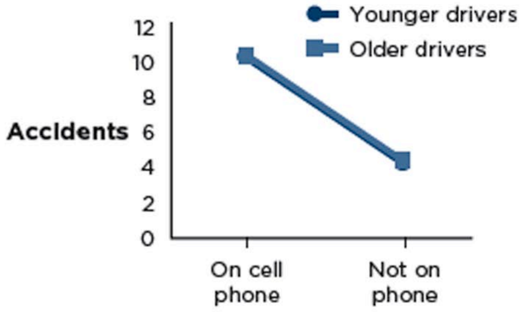

Main effect for cell phone: ***Yes***—cell phones cause more accidents

Main effect for age: ***Yes***—younger drivers have more accidents

Age × cell phone interaction: ***Yes***—for younger drivers, cell phones do not make a difference, but for older drivers, cell phones cause more accidents

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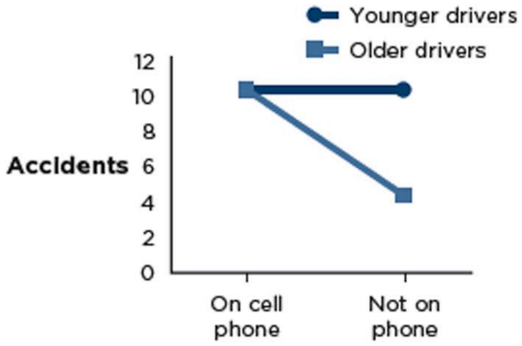
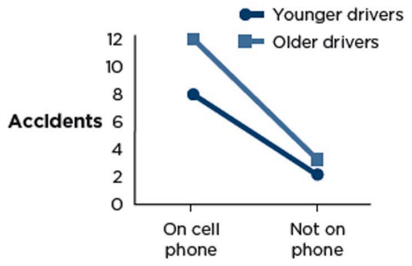


Possible Main Effects and Interactions in a 2 x 2 Factorial Design (continued)

Summary of effects | **Cell means and marginal means** | **Line graph of the results**

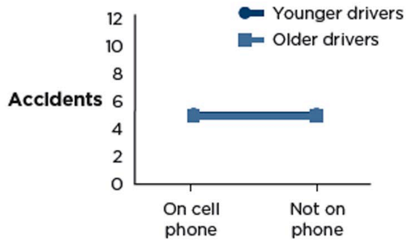
Main effect for cell phone:
Yes—cell phones cause more accidents
Main effect for age:
Yes—older drivers have more accidents
Age × cell phone interaction:
Yes—impact of a cell phone is larger for older than for younger drivers

DV: Number of accidents	On cell phone	Not on phone	
Younger drivers	8	2	5
Older drivers	12	3	7.5
	10	2.5	



Main effect for cell phone: ***No***
Main effect for age: ***No***
Age × cell phone interaction: ***No***

DV: Number of accidents	On cell phone	Not on phone	
Younger drivers	5	5	5
Older drivers	5	5	5
	5	5	



Main effect for cell phone:
Yes—cell phones cause more accidents
Main effect for age: ***No***
Age × cell phone interaction:
Yes—for younger drivers, cell phones make no difference, but for older drivers, cell phones cause more accidents

DV: Number of accidents	On cell phone	Not on phone	
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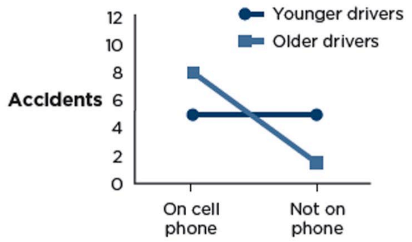
Older drivers	8	2	5
	6.5	3.5	



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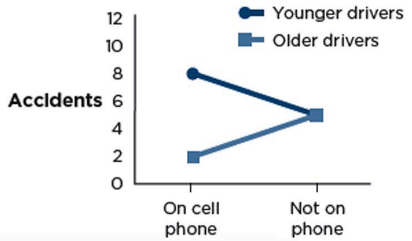

Possible Main Effects and Interactions in a 2 x 2 Factorial Design (continued)

Summary of effects | **Cell means and marginal means** | **Line graph of the results**

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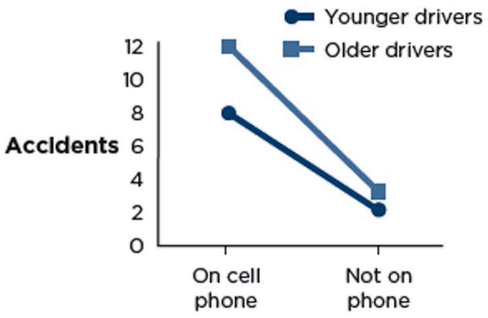

Main effect for cell phone: ***No***

Main effect for age: ***No***

Age × cell phone interaction: ***No***

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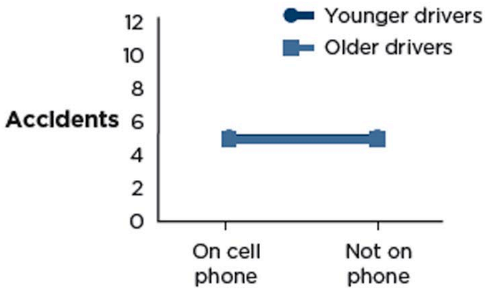
	5	5	



Possible Main Effects and Interactions in a 2 x 2 Factorial Design (continued)

Main effect for cell phone: ***Yes***—cell phones cause more accidents

Main effect for age: ***No***

Age × cell phone interaction: ***Yes***—for younger drivers, cell phones make no difference, but for older drivers, cell phones cause more accidents

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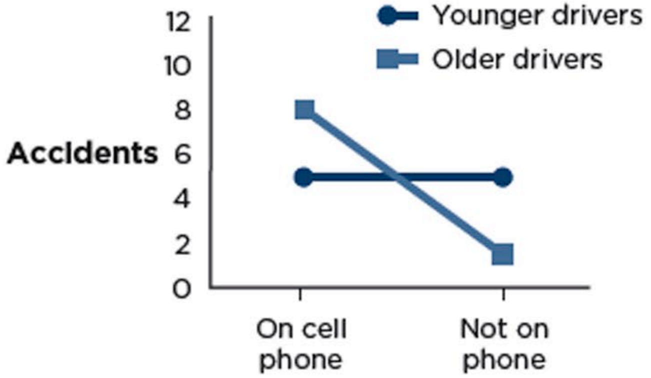
Older drivers	8	2	5
	6.5	3.5	



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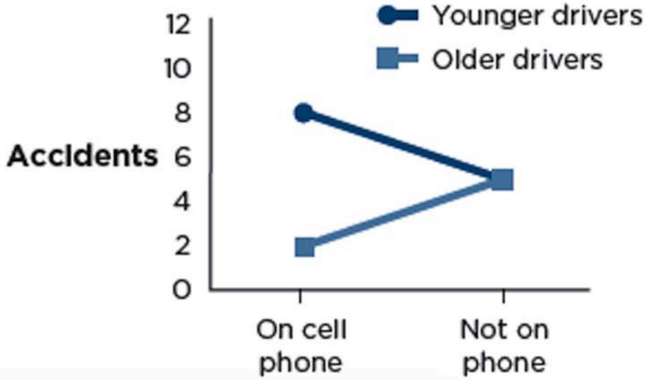


Wrapping Up…

Factorial Variations

This Time…

Factorial Variations

- Independent-groups factorial designs
- Within-groups factorial designs
- Mixed factorial designs
- Increasing the number of levels of an independent variable
- Increasing the number of independent variables

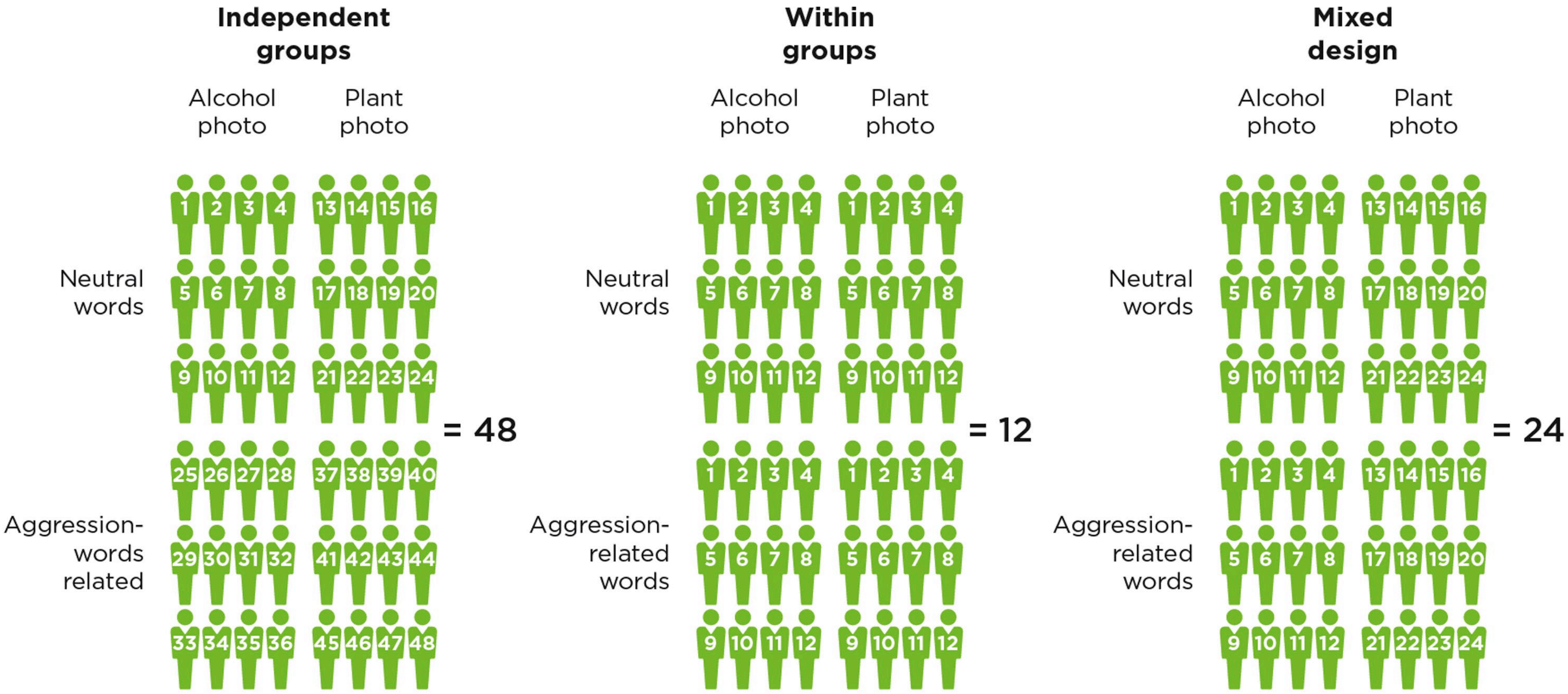
Independent-Groups Factorial Designs

Both IVs are studied as independent groups.

A 2 x 2 independent-groups factorial design has four groups or cells.

This is also known as a between-subjects factorial design.

Within-Groups Factorial Designs

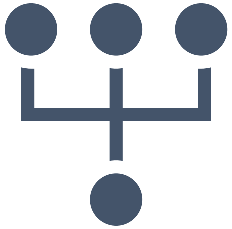


Mixed Factorial Designs

One IV is

manipulated as independent-groups

and the other is manipulated within-groups.



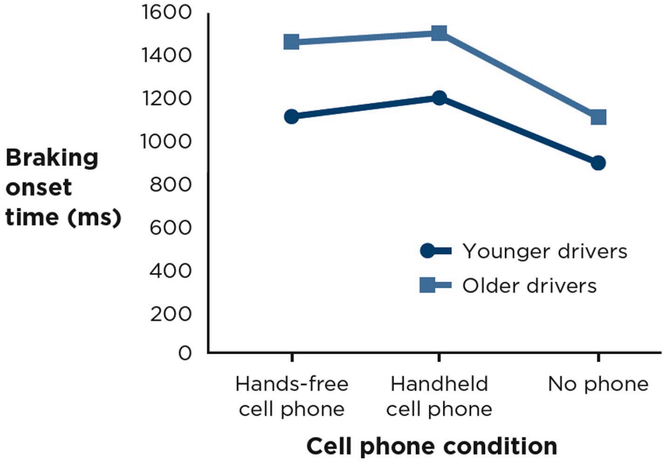
This design is intermediate

between the within-groups design and the independent-groups design

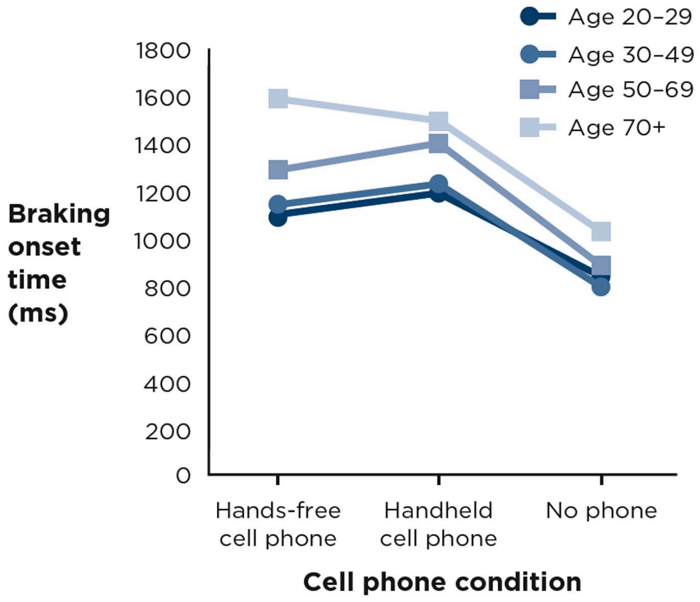
in terms of number of participants.

Increasing the Number of Levels of an Independent Variable

DV: Braking onset time (ms)		IV_1: Cell phone condition		
		Hands-free cell phone	**Handheld cell phone**	**No phone**
IV_2: Driver age	**Younger drivers**	1100	1200	850
	Older drivers	1450	1500	1050



DV: Braking onset time (ms)		IV_1: Cell phone condition		
		Hands-free cell phone	**Handheld cell phone**	**No phone**
IV_2: Driver age	**Age 20–29**	1100	1200	850
	Age 30–49	1120	1225	820
	Age 50–69	1300	1400	900
	Age 70+	1600	1500	1050



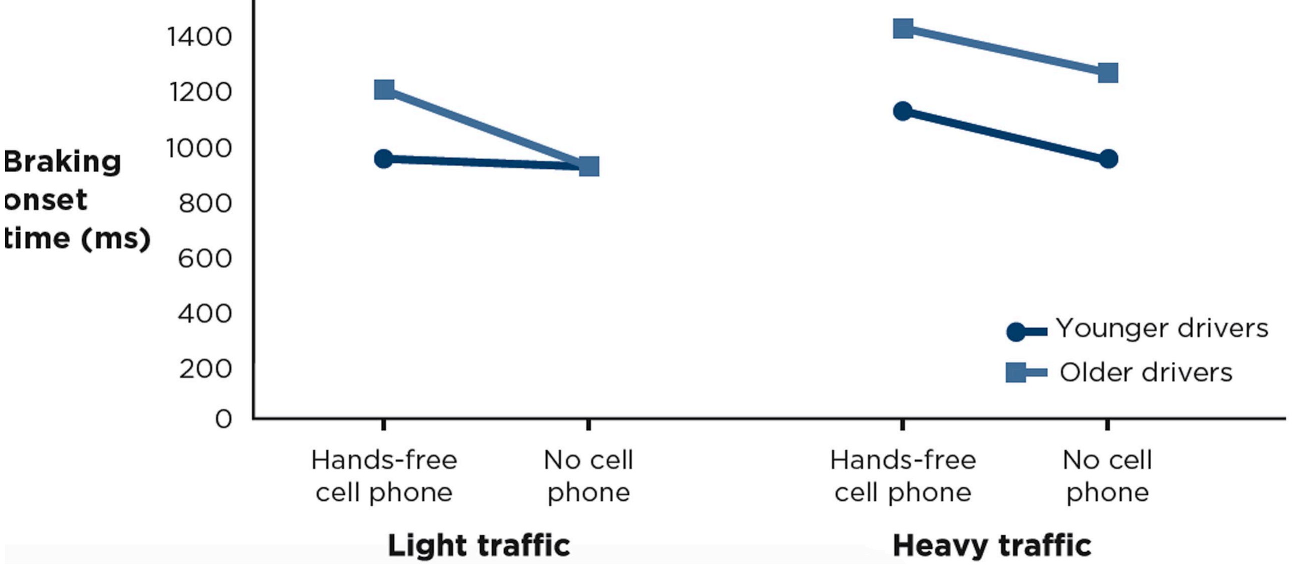
Increasing the Number of Independent Variables

- Main effects and interactions from a three-way design
- Why worry about all these interactions?

A

DV: Braking onset time	Light traffic			Heavy traffic		
		On cell phone	No cell phone		On cell phone	No cell phone
	Younger drivers	957	928	Younger drivers	1112	933
	Older drivers	1200	929	Older drivers	1412	1250

B



Main Effects and Interactions from a Three-Way Design

- Main effects: Is there a difference?

Main Effects in a Three-Way Factorial Design

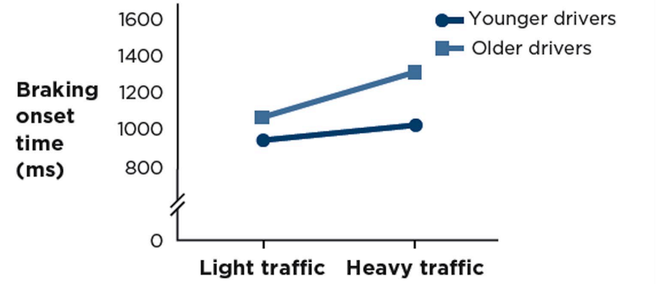
MAIN EFFECT FOR CELL PHONE VS. CONTROL	
Cell phone (mean of 957, 1200, 1112, 1412)	1170.25
No cell phone (mean of 928, 929, 933, 1250)	1010.00
MAIN EFFECT FOR DRIVER AGE	
Younger drivers (mean of 957, 928, 1112, 933)	982.50
Older drivers (mean of 1200, 929, 1412, 1250)	1197.75
MAIN EFFECT FOR TRAFFIC CONDITIONS	
Light traffic (mean of 957, 1200, 928, 929)	1003.50
Heavy traffic (mean of 1112, 1412, 933, 1250)	1176.75

Note: To estimate each main effect in a three-way factorial design, you average the means for braking onset time for each level of each independent variable, ignoring the levels of the other two independent variables. The means come from Figure 12.2. (These estimates assume there are equal numbers of participants in each cell.)

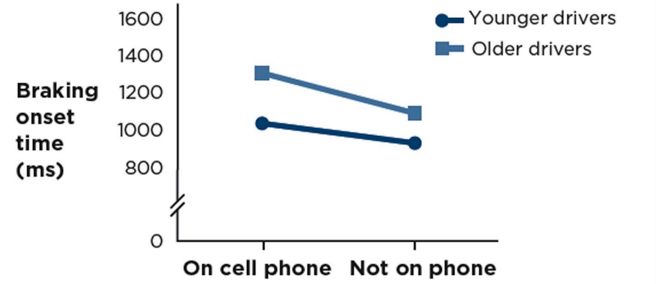
Main Effects and Interactions from a Three-Way Design: Two-Way Interaction

- Two-way interactions: Is there a difference in differences?

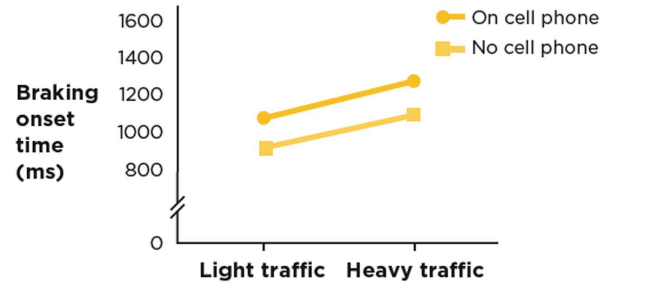
	Light traffic	Heavy traffic
Younger drivers	942.5 (average of 957 and 928)	1022.5 (average of 1112 and 933)
Older drivers	1064.5 (average of 1200 and 929)	1331 (average of 1412 and 1250)



	On cell phone	No cell phone
Younger drivers	1034.5 (average of 957 and 1112)	930.5 (average of 928 and 933)
Older drivers	1306 (average of 1200 and 1412)	1089.5 (average of 929 and 1250)



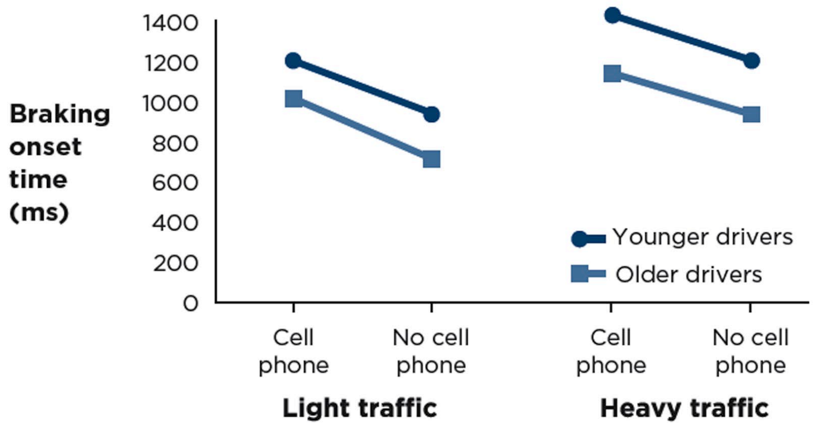
	Light traffic	Heavy traffic
On cell phone	1078.5 (average of 957 and 1200)	1262 (average of 1112 and 1412)
No cell phone	928.5 (average of 928 and 929)	1091.5 (average of 933 and 1250)



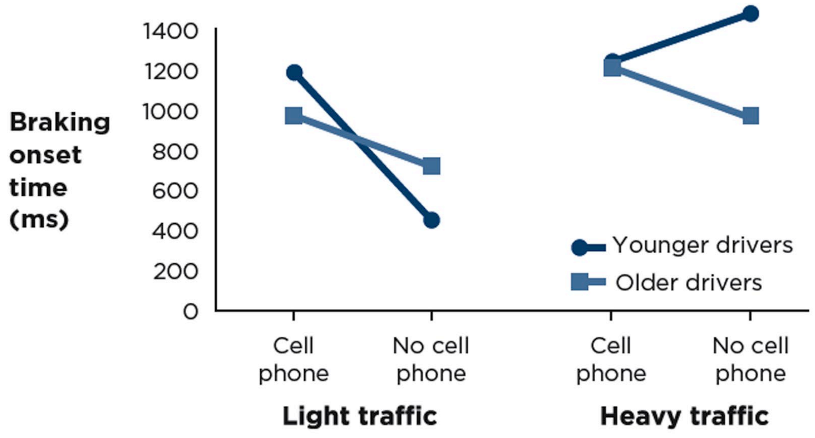
Three-way interactions: Are the two-way interactions different?

Main Effects and Interactions from a Three-Way Design: Three-Way Interaction

Possible outcome | **Is there a three-way interaction?**



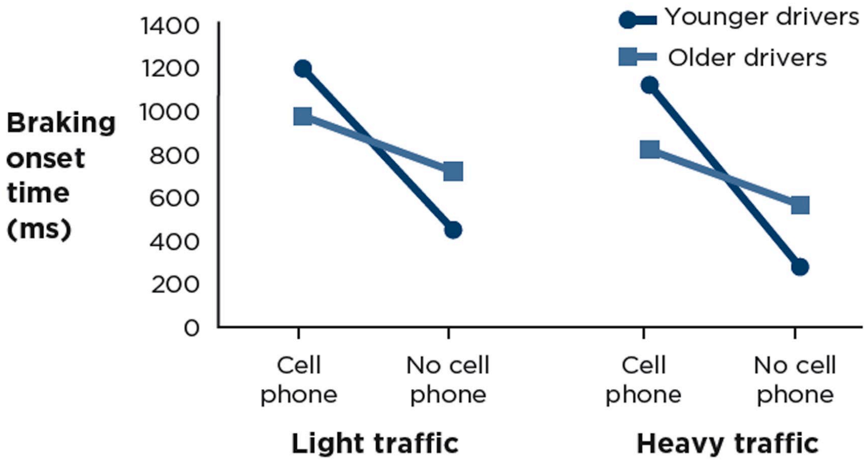
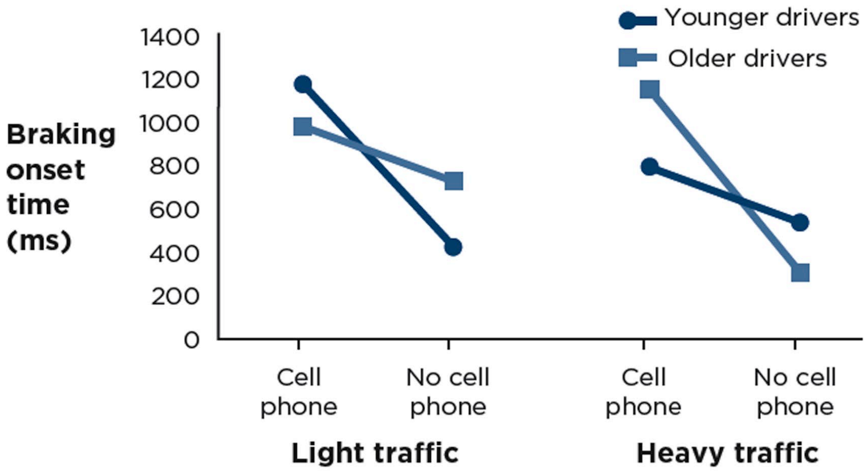
No.
There is no two-way interaction on the left side (light traffic) and no two-way interaction on the right side (heavy traffic). The two sides show the ***same*** two-way interaction (in both cases, the two-way interaction is zero, so there is no three-way interaction).



Yes.
There is a crossing interaction on the left side (light traffic), and a spreading interaction on the right side (heavy traffic). The two sides show different two-way interactions. So there is a three-way interaction.

Three-way interactions: Are the two-way interactions different?

Main Effects and Interactions from a Three-Way Design (continued)

Possible outcome	Is there a three-way interaction?
	***No.*** There is a two-way interaction on each side of the figure, and the two-way interaction is the same for both light traffic and heavy traffic. Therefore, the two sides show the same two-way interaction, so there is no three-way interaction.
	***Yes.*** Look carefully—the two-way interaction on the left side (light traffic) is actually the opposite of the two-way interaction on the right side (heavy traffic). So there is a three-way interaction.

Why Test All These Interactions?

Most outcomes in psychological studies are not main effects, but rather interactions.

Identifying Factorial Designs in Your Reading

IDENTIFYING FACTORIAL DESIGNS IN EMPIRICAL JOURNAL ARTICLES

IDENTIFYING FACTORIAL DESIGNS IN POPULAR PRESS ARTICLES

Identifying Factorial Designs in Empirical Journal Articles

- The Method section will describe the design of the study.
    - Factorial notation: _____ x _____ x _____

- The Results section will examine whether the main effects and interactions were significant.
    - Reference to "significance" or a *p* value and information about MANOVA and *F*

Identifying Factorial Designs in Popular Media Articles

- Look for phrases such as “it depends” or “only when” used to highlight an interaction.
- Look for participant variables (e.g., age, gender, ethnicity).

Wrapping Up…